

Daniel Akanji

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PROFESSIONAL SUMMARY

Robotics engineer and Ph.D. candidate with hands-on experience developing autonomous systems across robotic hardware, simulation, and experimental data. Proficient in Python, C++, ROS/ROS2, PyTorch, and HoloOcean, with expertise in state estimation, sensor fusion, planning, control, and multi-rate sensor analysis. Builds systems from CAD and electronics through software bring-up, simulation, field validation, and peer-reviewed publication. Research spans uncertainty-aware autonomy, predictive sensor scheduling, multi-robot coordination, and deployable marine robotic systems.

EDUCATION

Old Dominion University

Ph.D. in Mechanical Engineering.

Norfolk, VA

Aug 2024 - Present

Instituto Superior Tecnico & Universite de Toulon

Erasmus Mundus M.Sc. in Marine and Maritime Intelligent Robotics.

Portugal / France

Nov 2023

Federal University of Technology Akure

B.Eng. in Marine Science and Technology.

Nigeria

Aug 2017

EXPERIENCE

Graduate Researcher

Old Dominion University

Aug 2024 – Present

Norfolk, VA

- Led end-to-end development of BlueROV-based robotic systems spanning mechanical payload design, embedded integration, multi-sensor instrumentation, autonomy software, simulation, experimental planning, and deployment.
- Integrated Nortek Nucleus DVL/ADCP, CTD, IMU, cameras, and custom actuators with BlueOS and ArduSub; developed Python workflows for timestamp alignment, sensor-validity analysis, external-navigation evaluation, and synchronized log processing.
- Validated 5,108 of 6,175 bottom-track measurements (82.7%) during pool testing and traced failures to communication, frame alignment, range configuration, and estimator behavior, converting results into system-level configuration improvements.
- Built and validated a HoloOcean-based simulation and MATLAB replay pipeline: configured virtual DVL, IMU, and pose sensors; replayed real sensor data; compared simulation output against pool measurements; and tested navigation, control, uncertainty-aware planning, and sensor-scheduling methods before physical trials.
- Designed, fabricated, and tested ROV-mounted water and sediment sampling hardware using Fusion 360, PETG 3D printing, underwater servo actuation, linkages, and PyMAVLink control.

Research Intern

Institute for Systems and Robotics

Jan 2023 – Sep 2023

Lisbon, Portugal

- Implemented and integrated ROS-based C++ and Python software for an autonomous surface craft coordinating with two AUVs for ocean-observation missions.
- Developed particle-filter tracking, intermittent communication strategies, line-of-sight and pure-pursuit guidance, and inner-loop surge, pitch, and yaw control.
- Designed structured simulation campaigns, tuned controllers, and analyzed tracking and path-following performance across representative mission scenarios.

PROJECTS

Uncertainty-Aware Tracking of Multiple Drifting Patches

Python, MATLAB, state estimation, task scheduling, TV-LQR, environmental flow data

- Developed an integrated estimation, planning, and control framework for autonomous robots tracking multiple moving environmental targets in uncertain flow fields.
- Combined uncertainty propagation, service-aware task scheduling, and feedback control; validated the approach using high-frequency radar current replay and published results in *Frontiers in Robotics and AI*.

Predictive, Degradation-Aware Sonar Sensor Scheduling

Python, MATLAB, predictive optimization, underwater navigation, simulation

- Developed a predictive sensor-scheduling method for underwater navigation under time-varying sonar degradation and evaluated navigation performance through repeatable simulation experiments.
- Published the work at IEEE ROSE 2026, demonstrating research at the intersection of sensing, autonomy, and system reliability.

BlueROV Hardware, Simulation, and Multi-Sensor Data Validation

BlueROV Heavy, HoloOcean, DVL/ADCP, IMU, CTD, BlueOS, ArduSub, Python

- Created a hardware-to-simulation workflow for navigation-system development, including virtual sensor configuration, physical integration, synchronized multi-rate data analysis, and pool-based validation.
- Diagnosed estimator and sensor failures across hardware, communications, coordinate frames, and software configuration, documenting reproducible fixes and test procedures.

Multi-Robot Ocean Observation and River-Flow Estimation

ROS, C++, Python, MATLAB, particle filtering, support-vector regression, field robotics

- Developed cooperative tracking and control software for an autonomous surface craft and two AUVs; extended multi-robot field experience through three Elizabeth River trials using three aquatic robots.
- Built data-processing and machine-learning workflows to estimate river-current speed and direction from robot-collected measurements.

Reinforcement Learning Environment Development

Python, Gymnasium, PyTorch, PPO, simulation and evaluation

- Built a custom Gymnasium-compatible multi-agent decision environment and trained and evaluated PPO agents, deepening experience in simulation construction, reward design, and reproducible learning experiments.

SKILLS

Programming & ML: Python, C++, MATLAB, Bash, PyTorch, Gymnasium, reinforcement learning (PPO), NumPy, Pandas

Robotics & Simulation: ROS/ROS2, HoloOcean, MATLAB/Simulink, BlueOS, ArduSub, MAVLink/PyMAVLink, Monte Carlo simulation

Autonomy & Data: state estimation, sensor fusion, motion control, path planning, multi-robot systems, time synchronization, data validation, log analysis

Hardware & Prototyping: BlueROV Heavy, Raspberry Pi, Arduino, DVL/ADCP, IMU, CTD, sonar, cameras, servo actuation, Fusion 360, 3D printing

Tools: Linux, Git, Jupyter, scientific computing, experimental design, technical writing

SELECTED PUBLICATIONS

- Uncertainty-aware estimation, planning, and control for tracking multiple drifting patches in flow fields. *Frontiers in Robotics and AI*, 2026. DOI: 10.3389/frobt.2026.1842384
- Predictive, Degradation-Aware Sensor Scheduling for Underwater Sonar Navigation. *IEEE International Symposium on Robotic and Sensors Environments (ROSE)*, 2026. DOI: 10.1109/ROSE69181.2026.11570985
- Passive Attitude Biasing for Low-Disturbance Sediment Sampling in PFAS Studies Using an Observation-Class ROV. *OCEANS 2026 Monterey* - accepted for presentation and publication.

SELECTED EXPERIMENTAL & DATA EXPERIENCE

- Processed and aligned multi-rate robotic datasets, including 100 Hz IMU streams, 10 Hz AHRS/INS outputs, DVL bottom-track measurements, ArduPilot logs, and CTD records.
- Designed repeatable bench, tank, pool, simulation, and river-test procedures; translated observed failures into mechanical, sensor, configuration, and software revisions.
- Built Jupyter/Python analysis workflows for timestamp alignment, coordinate-frame checks, measurement validity, trajectory comparison, sensor quality, and controller performance.
- Created portfolio-ready technical artifacts including CAD renders, system architecture diagrams, validation plots, experiment photographs, and short robot demonstration videos.